FAITHFULLY FLAT DESCENT FOR THE PICARD GROUP WHEN THE RING OF GLOBAL SECTIONS IS SEMI-LOCAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites three cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

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Allowed standard prerequisites (editorial).

- Invertible modules over a semi-local ring are free of rank one. Stacks Project, Algebra, Lemma algebra-lemma-locally-free-semi-local-free (tag 00NX). Turns the invertible section modules into copies of R.
- The faithful flat descent statement for line bundles with trivial pullback. Stacks Project, Varieties, Lemma lemma-change-rings-pic-pre (tag 0C4H). Supplies both the invertibility of the section modules and the unit property of st.

Source reference. Original lemma 0CDX: sources/stacks-chow/tags.txt, source lines 15557–15557. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02M9: sources/stacks-chow/tags.txt, source lines 3436–3436. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $A \to B$ be a faithfully flat ring map. Let X be a scheme over A such that*

(1) *X is quasi-compact and quasi-separated, and*
(2) *$R = H^0(X, \mathcal{O}_X)$ is a semi-local ring.*

Then the pullback map $\mathrm{Pic}(X) \to \mathrm{Pic}(X_B)$ is injective.

Proof. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module whose pullback $\mathcal{L}'$ to X_B is trivial. Set $M = H^0(X, \mathcal{L})$ and $N = H^0(X, \mathcal{L}^{\otimes -1})$. By Lemma 0CDX the R-modules M and N are invertible. Since R is semi-local $M \cong R$ and $N \cong R$, see Algebra, Lemma 02M9. Choose generators $s \in M$ and $t \in N$. Then $st \in R = H^0(X, \mathcal{O}_X)$ is a unit by the last part of Lemma 0CDX. We conclude that s and t define trivializations of $\mathcal{L}$ and $\mathcal{L}^{\otimes -1}$ over X. □

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